

IT3021

Data Warehousing and Business Intelligence



Assignment 02

Implementation of a Business Intelligence Solution using SSAS, OLAP, and Power BI for US Regional Sales Data Warehouse

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1 Introduction

This assignment focuses on developing a Business Intelligence (BI) solution using the data warehouse created in Assignment 1 for US Regional Sales Analysis. The objective is to analyze sales data and generate meaningful insights using analytical and visualization tools.

An SSAS cube is implemented to enable multidimensional analysis by defining measures and dimensions with hierarchies. Microsoft Excel is then used to perform OLAP operations such as roll-up, drill-down, slice, dice, and pivot using Pivot Tables. Finally, Power BI is used to create interactive reports and dashboards for better data visualization and decision-making.

Overall, this assignment demonstrates how a data warehouse can be transformed into a powerful BI solution using the Microsoft BI stack.

2 STEP 1: Data Source

2.1 Overview of Data Warehouse

The data warehouse used in this assignment was developed in Assignment 1 for the **US Regional Sales Analysis** domain. It is designed using a **star schema** to support analytical queries and reporting.

The data warehouse integrates sales-related data from operational sources and stores it in a structured format suitable for Business Intelligence and decision-making.

2.2 Schema Description

The data warehouse consists of one fact table and several dimension tables:

- **Fact Table:**
 - **FactSales** – Stores transactional sales data including measures such as sales amount and quantity.
- **Dimension Tables:**
 - **DimCustomer** – Contains customer-related information.
 - **DimDate** – Stores date attributes such as year, quarter, and month.
 - **DimProduct** – Includes product-related details.
 - **DimSalesChannel** – Represents the sales channel (e.g., online, offline).
 - **DimStore** – Contains store or location-related information.

These dimension tables provide descriptive attributes that allow detailed analysis of sales data from multiple perspectives.

2.3 Schema Diagram



2.4 Key Features of DW

- The data warehouse is structured using a **star schema design**.
- It includes key **measures** such as sales amount and quantity in the FactSales table.
- Multiple dimensions enable **multi-dimensional analysis** (customer, product, store, time, and sales channel).
- The schema is optimized for use with **SSAS, Excel OLAP operations, and Power BI reporting**.

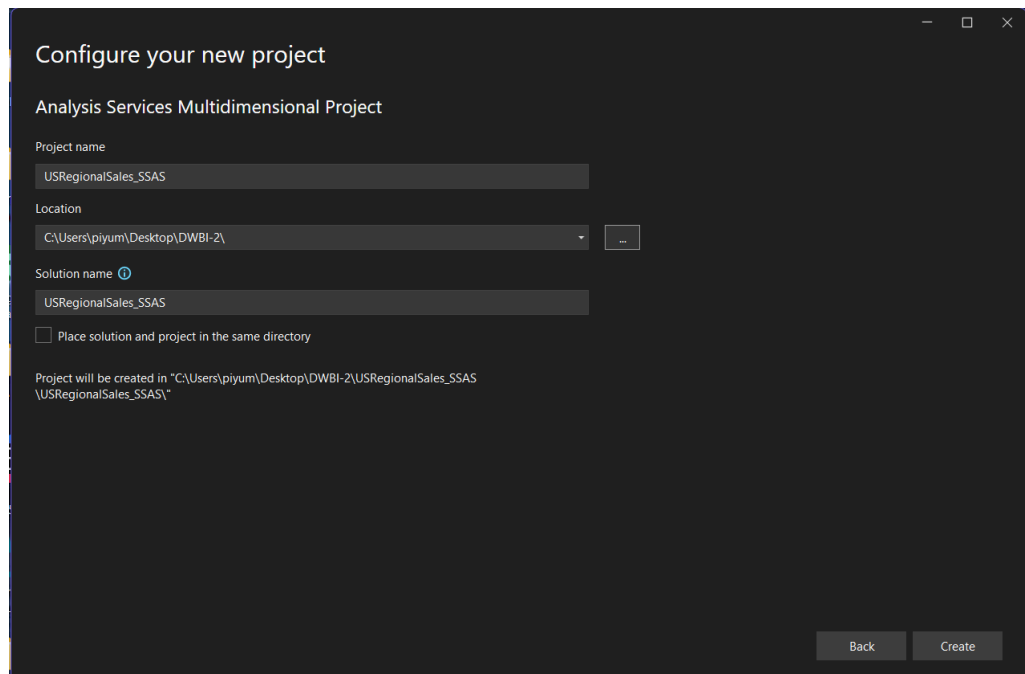
3 STEP 2: SSAS Cube Implementation

3.1 Introduction to SSAS Cube

An SSAS (SQL Server Analysis Services) cube is used to enable **multidimensional analysis** of data stored in the data warehouse. It organizes data into measures and dimensions, allowing fast querying and aggregation.

In this assignment, an SSAS cube is created using the **USRegionalSales_DW** database to support analytical operations.

3.2 Creating SSAS Project



3.3 Data Source & Data Source View

A data source connection was created to connect the SSAS project with the USRegionalSales_DW database using SQL Server.

Data Source Wizard

Select how to define the connection
You can select from a number of ways in which your data source will define its

☒ Create a data source based on another object

☐ Create a data source based on an existing or new connection

Data connections:

Property	Value
Data Source	LAPTOP-2P5598FL\MSS...
Initial Catalog	USRegionalSales_DW
Provider	SQLNCLI11.1
User ID	sa

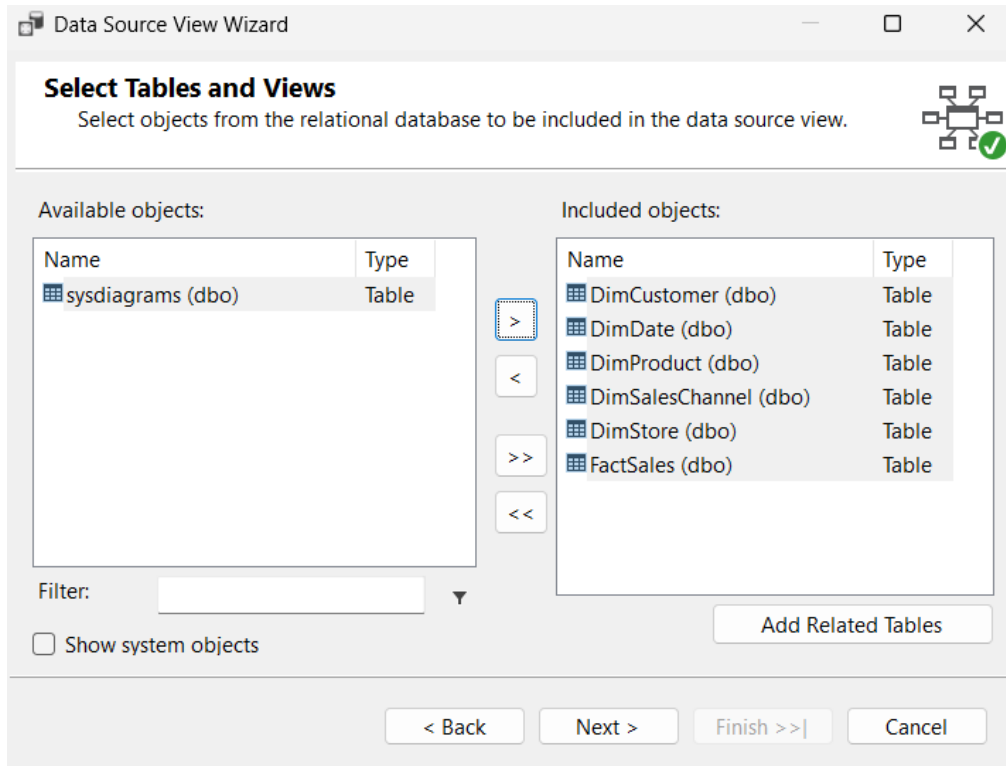
New... Delete

< Back Next > Finish >>| Cancel

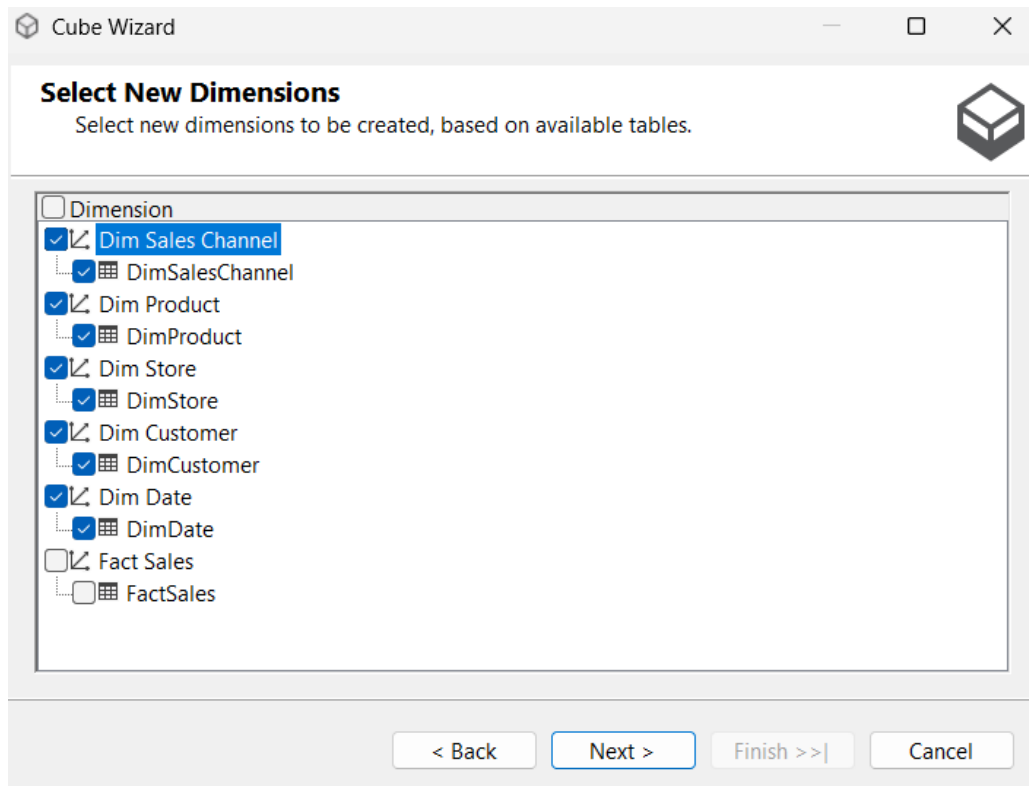
A Data Source View was created by selecting the required tables:

- FactSales
- DimCustomer
- DimDate
- DimProduct
- DimSalesChannel
- DimStore

Relationships between tables were automatically detected.



3.4 Creating Dimensions



3.5 Creating Measures (Fact Table)

Cube Wizard

Select Measure Group Tables
Select a data source view or diagram and then select the tables that will be used for measure groups.

Data source view:
US Regional Sales DW

Measure group tables:

- ☐ DimCustomer
- ☐ DimDate
- ☐ DimProduct
- ☐ DimSalesChannel
- ☐ DimStore
- ☒ FactSales

Suggest

< Back Next > Finish >>| Cancel

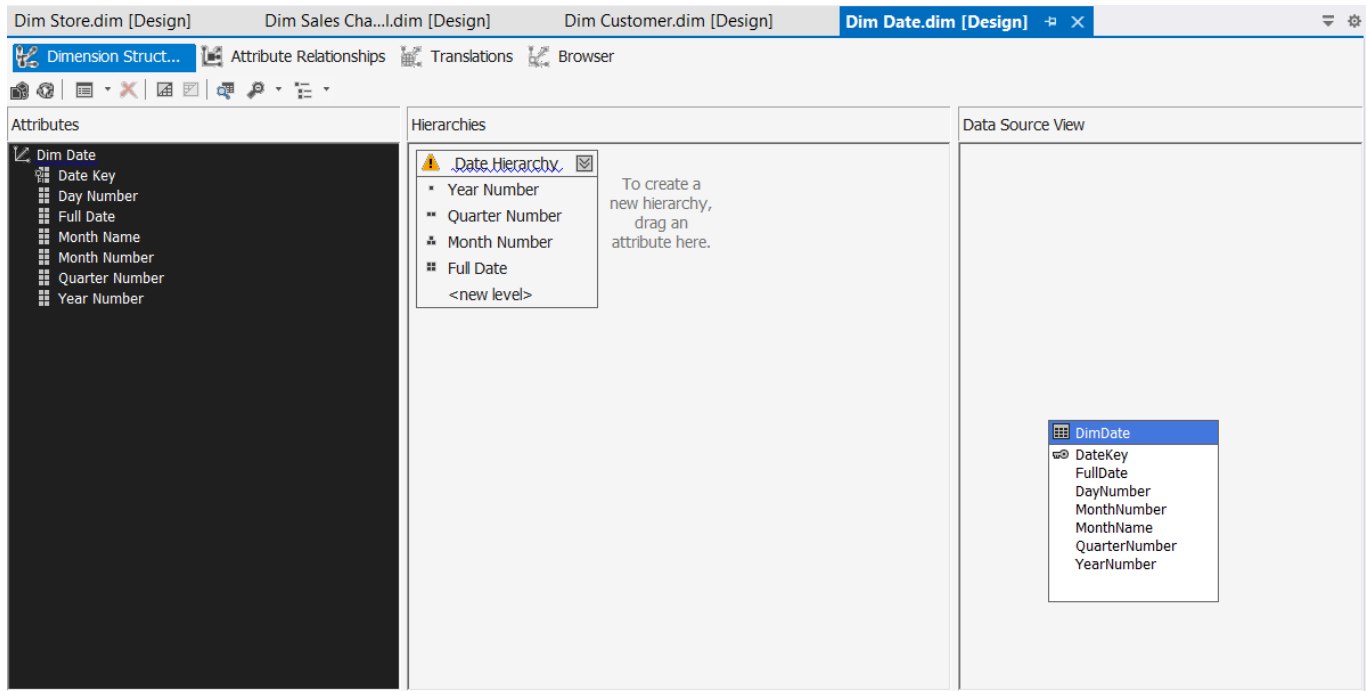
3.6 Creating Hierarchies

Hierarchies were created to support drill-down analysis.

For example:

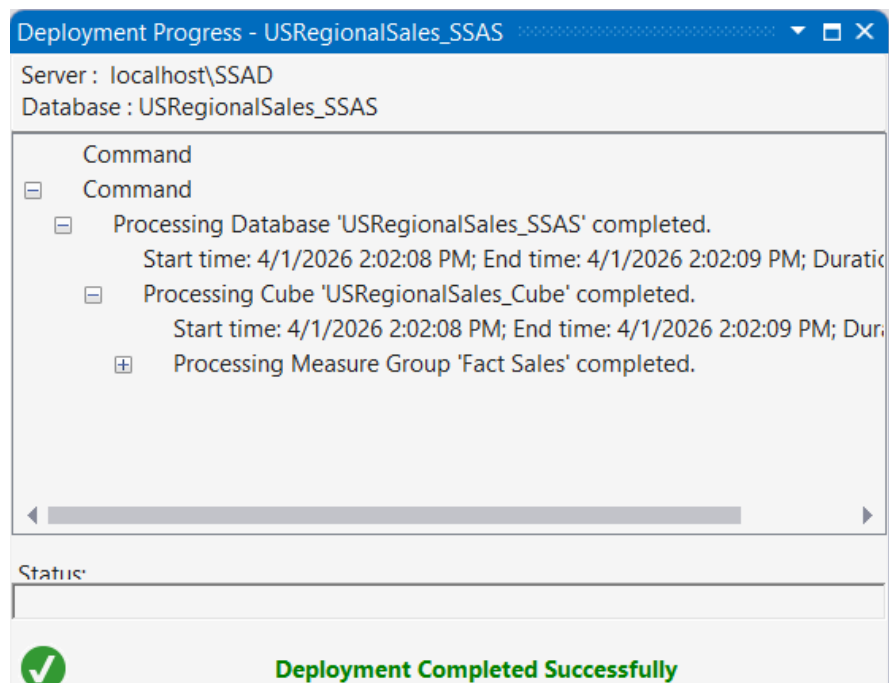
- **Date Hierarchy:**
Year → Quarter → Month

This allows users to analyze data at different levels of detail.



3.7 Cube Deployment & Processing

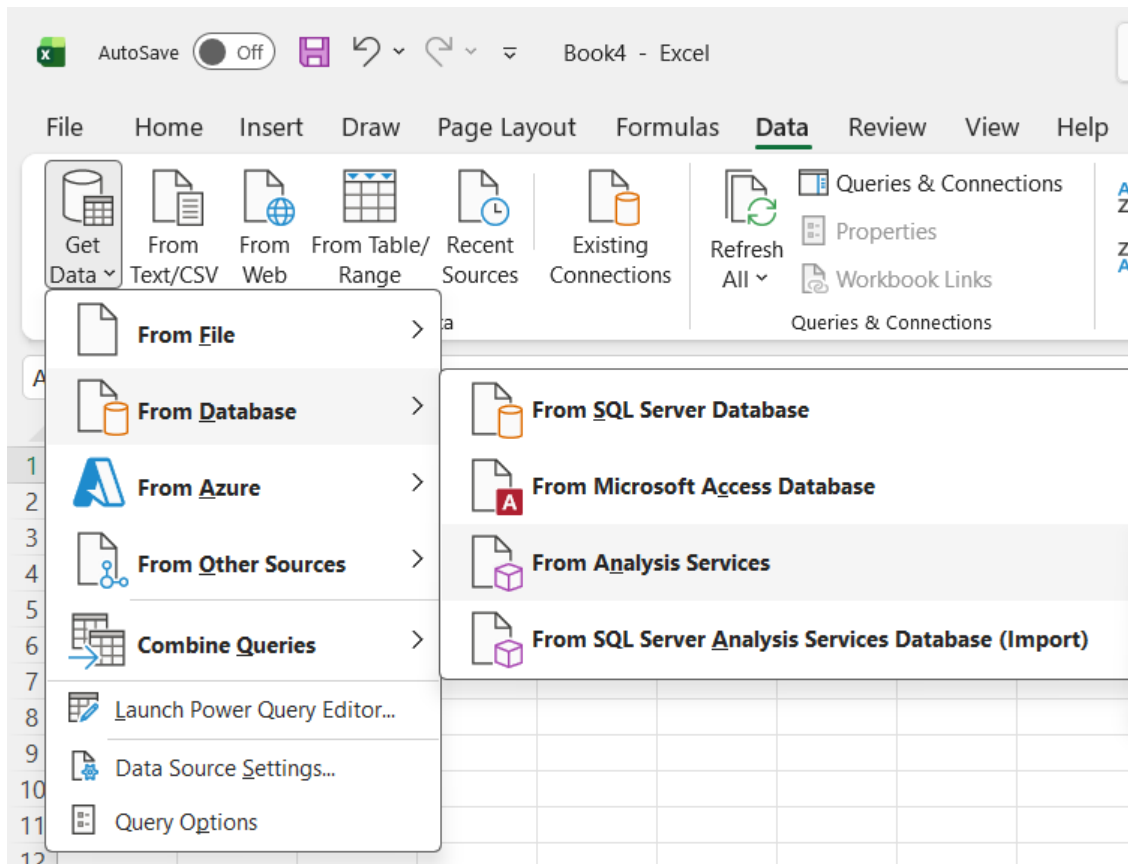
The cube was deployed and processed successfully. Data was loaded into the cube and verified using the browser tab.



4 STEP 3: OLAP Operations using Excel

4.1 Connecting Excel to SSAS Cube

Microsoft Excel was connected to the deployed SSAS cube using the “From Analysis Services” option. The cube was selected, and a Pivot Table was created to perform OLAP analysis.



4.2 Creating Pivot Table

A Pivot Table was created by dragging dimensions and measures into Rows, Columns, and Values areas to analyze sales data interactively.

PivotTable Fields

Choose fields to add to report:

Search

Fact Sales

☐ Cost Amount

☐ Discount Applied

☐ Fact Sales Count

☐ Order Quantity

☒ Sales Amount

☐ Type Process Time Hours

Drag fields between areas below:

Filters

Columns

Date Hierarchy

Rows

Store Name

Values

Sales Amount

☐ Defer Layout Update

Update

4.3 OLAP Operation 1 – Roll-up

The Roll-up operation aggregates data to a higher level. Sales data was summarized at the **Year level** using the Date hierarchy, providing an overview of total sales per year.

	A	B	C	D	E	F	G	
1								
2								
3	Sales Amount	Column Labels						
4	Row Labels	Store 001	Store 002	Store 003	Store 004	Store 005	Store 006	Store 007
5	2018	37801.4	120291.8	46953.6	30578.8	27630.8	151393.2	6
6	2019	235076.2	137363.4	82088.4	144331.4	56079	98047.8	14
7	Grand Total	272877.6	257655.2	129042	174910.2	83709.8	249441	20
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								
19								

PivotTable Fields

Choose fields to add to report:

Fact Sales

☐ Cost Amount

☐ Discount Applied

☐ Fact Sales Count

☐ Order Quantity

☒ Sales Amount

☐ Time Process Time Hours

Drag fields between areas below:

Filters

4.4 OLAP Operation 2 – Drill-down

The Drill-down operation allows viewing detailed data. The Date hierarchy was expanded from **Year to Quarter and Month levels**, enabling detailed analysis of sales.

	A	B	C	D	E	F	G
1							
2	Sales Amount	Column Labels					
3	Row Labels	Store 001	Store 002	Store 003	Store 004	Store 005	Store 006
4	2018						
5	2						
6	5						
7	2018-05-31						9540
8	6	4998.2					328
9	3						
10	7	6472.2	31570.4		8844		18130
11	8	22177			8951.2	24414.8	14720
12	9			46953.6			
13	4	4154	88721.4		12783.6	3216	7616
14	2019						
15	1						
16	1						
17	2019-01-01						
18	2019-01-02						
19	2019-01-03						
20	2019-01-04						
21	2019-01-05						
22	2019-01-06						
23	2019-01-07						
24	2019-01-08						
25	2019-01-09						
26	2019-01-10		10653				
27	2019-01-11						
28	2019-01-12						

PivotTable Fields

Choose fields to add to report:

Search

Fact Sales

- ☐ Cost Amount
- ☐ Discount Applied
- ☐ Fact Sales Count
- ☐ Order Quantity
- ☒ Sales Amount
- ☐ Time Process Time Hours

Drag fields between areas below:

Filters

Rows

Date Hierarchy

4.5 OLAP Operation 3 – Slice

The Slice operation filters data based on a single dimension. For example, sales data was filtered for a specific **year or category**, showing a subset of the data.

	A	B	C	D	E	F	G
1	Sales Channel Name	Online					
2							
3	Sales Amount	Column Labels					
4	Row Labels	Store 001	Store 002	Store 004	Store 005	Store 006	Store 007
5	2018						
6	2						
7	5						
8	2018-05-31						
9	6	4998.2				14257.6	
10	3						
11	7		31570.4	8844		18130.2	1510
12	8	22177		8951.2		14726.6	
13	9						
14	4					31985.8	908
15	2019	87394.8	82972.8	61532.8	10653	65847.6	297
16	Grand Total	114570	114543.2	79328	10653	144947.8	539
17							
18							
19							
20							

PivotTable Fields
Choose fields to add to report:
Search
Fact Sales
☐ Cost Amount
☐ Discount Applied
☐ Fact Sales Count
☐ Order Quantity
☒ Sales Amount
☐ Time Process Time Hours
Drag fields between areas below:
Filters
Sales Channel Name

4.6 OLAP Operation 4 – Dice

The Dice operation filters data using multiple dimensions. Sales data was analyzed by applying filters such as **specific year and product category**, providing a more refined view.

	A	B	C	D	E	F
3						
4						
5	Sales Amount	Column Labels				
6	Row Labels	2018	2019	Grand Total		
7	Store 001	37801.4	235076.2	272877.6		
8	Store 002	120291.8	137363.4	257655.2		
9	Store 003	46953.6	82088.4	129042		
10	Store 004	30578.8	144331.4	174910.2		
11	Store 005	27630.8	56079	83709.8		
12	Store 006	151393.2	98047.8	249441		
13	Store 007	64467.4	143353.2	207820.6		
14	Store 008	4422	147453.6	151875.6		
15	Store 009	87421.6	100406.2	187827.8		
16	Store 010	10291.2	109089.4	119380.6		
17	Store 011	65646.6	229247.2	294893.8		
18	Store 012	102255.4	278880.8	381136.2		
19	Store 013	26599	90664.4	117263.4		
20	Store 014	45600.2	166374.4	211974.6		
21	Store 015	38793	149785.2	188578.2		
22	Store 016	52796	180699	233495		
23	Store 017	227103.2	245086	472189.2		
24	Store 018	185268.4	11818.8	197087.2		
25	Store 019	141303	38310.6	179613.6		
26	Store 020	215271	88198.8	303469.8		
27	Store 021	183513	274606.2	458119.2		
28	Store 022	137095.4	223083.2	360178.6		
29	Store 023	116727.4	42518.2	159245.6		
30	Store 024	137363.4	233887	371250.4		

PivotTable Fields
Choose fields to add to report:
Search

Fact Sales
☐ Cost Amount
☐ Discount Applied
☐ Fact Sales Count
☐ Order Quantity
☒ Sales Amount
☐ Time Periods Time Hours

Drag fields between areas below:

Filters

Rows
Store Name

4.7 OLAP Operation 5 – Pivot

The Pivot operation changes the orientation of the data. Rows and columns were rearranged in the Pivot Table to view the data from a different perspective.

A5							
	A	B	C	D	E	F	G
1							
2							
3	Sales Amount	Column Labels					
4	Row Labels	Store 001	Store 002	Store 003	Store 004	Store 005	Store 006
5	2018	37801.4	120291.8	46953.6	30578.8	27630.8	151393.2
6	2019	235076.2	137363.4	82088.4	144331.4	56079	98047.8
7	Grand Total	272877.6	257655.2	129042	174910.2	83709.8	249441
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							

PivotTable Fields

Choose fields to add to report:

Search

Fact Sales

- ☐ Cost Amount
- ☐ Discount Applied
- ☐ Fact Sales Count
- ☐ Order Quantity
- ☒ Sales Amount
- ☐ Two Process Time Hours

Drag fields between areas below:

Filters

5 STEP 4: Power BI Reports

5.1 Data Loading into Power BI

SQL Server database

Server ①

localhost\SSAD

Database (optional)

USRegionalSales_SSAS

Data Connectivity mode ①

- ☒ Import
- ☐ DirectQuery

Advanced options

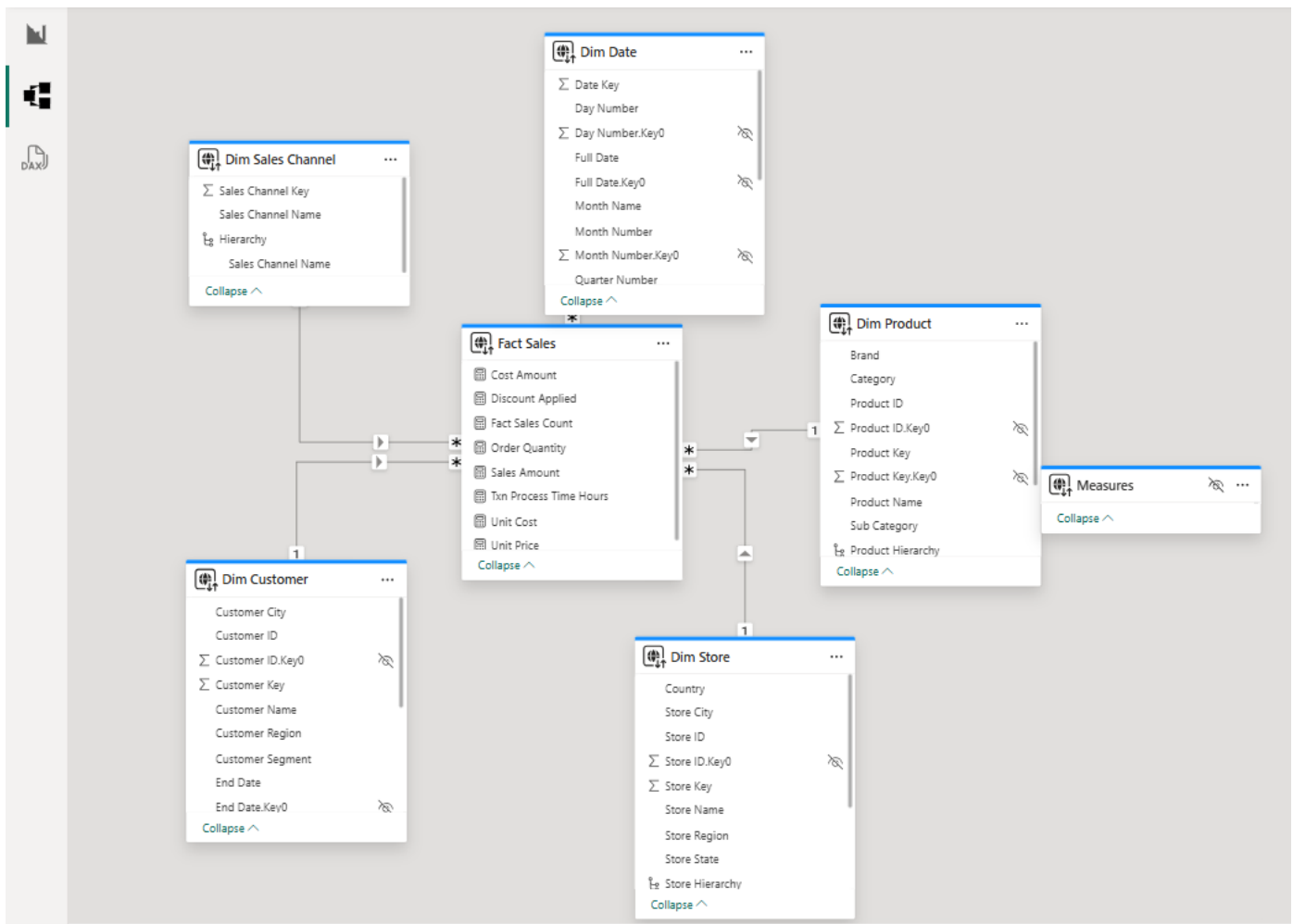
OK

Cancel

Power BI was connected to the data source (SQL Server / SSAS cube). The required tables/measures were loaded into Power BI for report creation.

5.2 Data Modeling

Relationships between tables were verified in the model view. The star schema structure ensures proper filtering and interaction between visuals.



5.3 Report 1 – Matrix Report

A Matrix visual was created to display sales data by dimensions such as **Year and Product Category**, with measures like Sales Amount.

Sales Analysis by Store and Date											
Year Number	Store 001	Store 002	Store 003	Store 004	Store 005	Store 006	Store 007	Store 008	Store 009	Store 010	Store 011
2018	37,801.40	120,291.80	46,953.60	30,578.80	27,630.80	151,393.20	64,467.40	4,422.00	87,421.60	10,291.20	65,646.40
2	4,998.20					42,370.80	20,984.40		43,576.80		15,972.00
June	4,998.20					32,830.00	20,984.40		43,576.80		15,972.00
May						9,540.80					
3	28,649.20	31,570.40	46,953.60	17,795.20	24,414.80	32,856.80	34,397.80				26,411.20
August	22,177.00			8,951.20	24,414.80	14,726.60					24,066.40
July	6,472.20	31,570.40		8,844.00		18,130.20	34,397.80				2,345.60
September			46,953.60								
4	4,154.00	88,721.40		12,783.60	3,216.00	76,165.60	9,085.20	4,422.00	43,844.80	10,291.20	23,262.40
December	4,154.00	2,318.20				44,179.80			43,844.80	10,291.20	
November		55,435.80		12,783.60		11,725.00					23,262.40
October		30,967.40			3,216.00	20,260.80	9,085.20	4,422.00			
2019	235,076.20	137,363.40	82,088.40	144,331.40	56,079.00	98,047.80	143,353.20	147,453.60	100,406.20	109,089.40	229,247.20
1	7,999.80	12,663.00		31,945.60		12,488.80	42,156.40	11,457.00			53,144.00
February				31,945.60		12,488.80		11,457.00			48,400.00
January	7,999.80	10,653.00					27,845.20				4,743.20
March		2,010.00					14,311.20				
2	66,812.40	8,656.40	15,356.40	34,786.40	37,185.00	66,933.00	22,914.00	21,198.80	46,323.80		89,257.20
April	57,405.60	8,656.40	15,356.40			27,054.60					82,280.00
Total	272,877.60	257,655.20	129,042.00	174,910.20	83,709.80	249,441.00	207,820.60	151,875.60	187,827.80	119,380.60	294,893.20

The matrix report provides a summarized view of sales data across multiple dimensions, allowing easy comparison of values.

5.4 Report 2 – Slicers (Cascading Filters)

Slicers were created using fields such as **Year and Sales Channel**. These slicers filter multiple visuals simultaneously.

The slicers allow users to interactively filter the data. Cascading filters ensure that selecting one slicer dynamically affects other visuals.

Year Number

☐ 2017

☐ 2018

☐ 2019

Month Name

☐ April

☐ August

☐ December

Store Name

☐ Store 001

☐ Store 002

☐ Store 003

Category

☐ Electronics

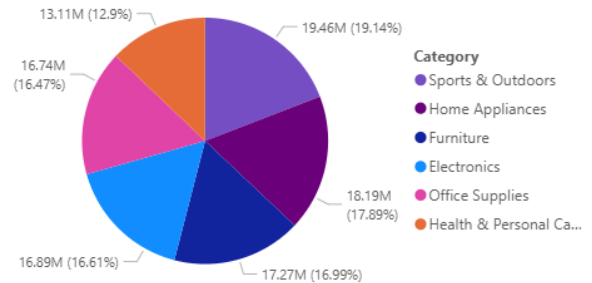
☐ Furniture

☐ Health & Personal Care

Sales Analysis by Store and Date

Year Number	Store 001	Store 002	Store 003	Store 004	Store 005	S
2018	37,801.40	120,291.80	46,953.60	30,578.80	27,630.80	1
2	4,998.20					
June	4,998.20					
May						
3	28,649.20	31,570.40	46,953.60	17,795.20	24,414.80	
August	22,177.00			8,951.20	24,414.80	
July	6,472.20	31,570.40		8,844.00		
September			46,953.60			
4	4,154.00	88,721.40		12,783.60	3,216.00	
December	4,154.00	2,318.20				
November		55,435.80		12,783.60		
October		30,967.40			3,216.00	
2019	235,076.20	137,363.40	82,088.40	144,331.40	56,079.00	
Total	272,877.60	257,655.20	129,042.00	174,910.20	83,709.80	2

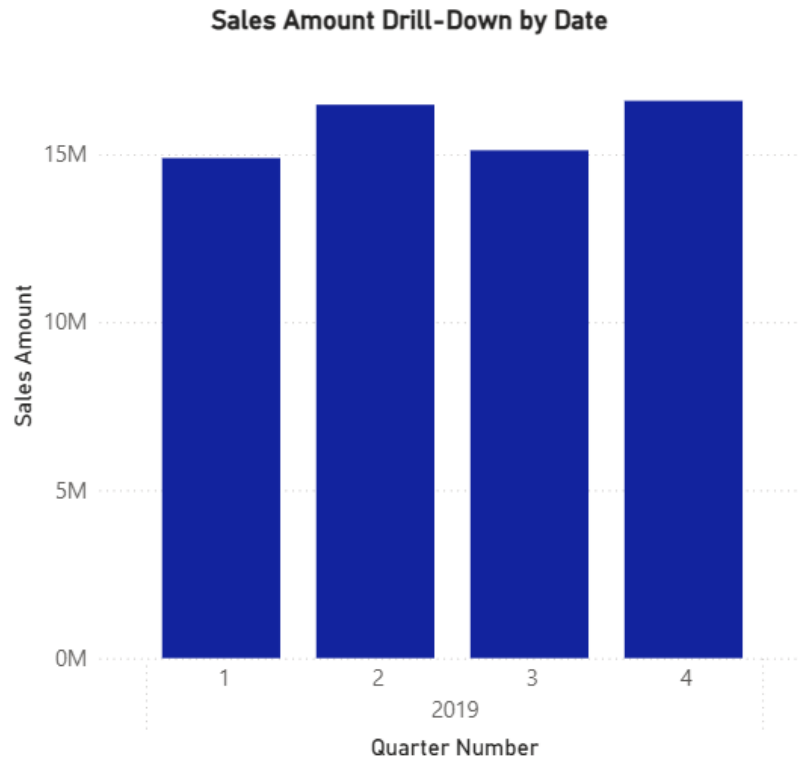
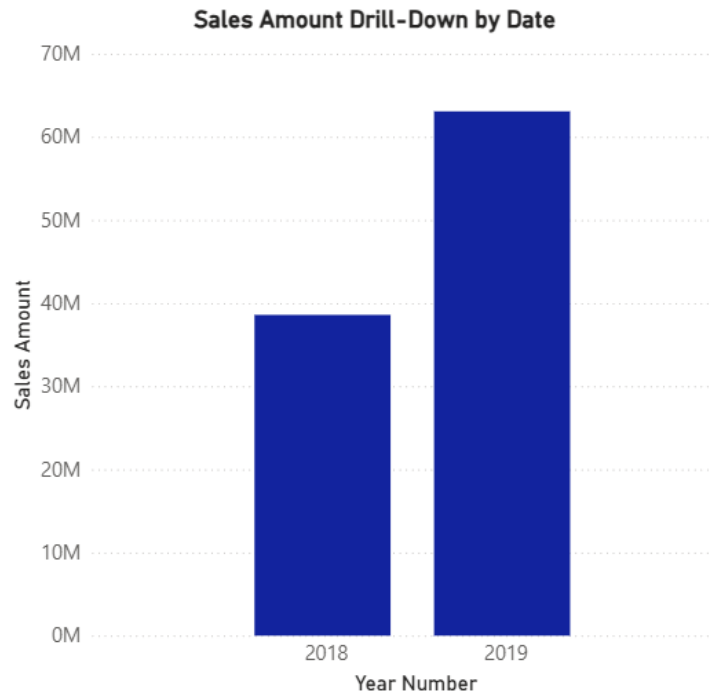
Sales Amount by Category



5.5 Report 3 – Drill-down Report

A chart was created using a hierarchy (e.g., **Year** → **Quarter** → **Month**) to enable drill-down functionality.

The drill-down feature allows users to explore data at different levels of detail, from summarized to detailed views.





Year Number	Month Name	Sales Amount	Order Quantity	Category	Customer City
2019	April	17,781.80	20	Electronics	Chicago
2019	April	19,872.20	14	Electronics	Dallas
2019	April	111,447.80	70	Electronics	Houston
2019	April	32,347.60	26	Electronics	Los Angeles
2019	April	74,356.60	42	Electronics	New York
2019	April	49,432.60	28	Electronics	Philadelphia
2019	April	46,618.60	46	Electronics	Phoenix
2019	April	120,466.00	52	Electronics	San Antonio
2019	April	156,123.40	58	Electronics	San Diego
2019	April	152,492.00	52	Electronics	San Jose
2019	April	87,528.80	38	Furniture	Chicago
2019	April	64,293.20	38	Furniture	Dallas
2019	April	44,179.80	32	Furniture	Houston
2019	April	139,132.20	52	Furniture	Los Angeles
2019	April	97,739.60	32	Furniture	New York
2019	April	46,819.60	34	Furniture	Philadelphia
2019	April	105,431.20	32	Furniture	Phoenix
2019	April	111,675.60	90	Furniture	San Antonio
2019	April	245,528.20	62	Furniture	San Diego
Total		63,069,579.00	27274		

6 Conclusion

This assignment successfully demonstrated the implementation of a complete Business Intelligence solution using the US Regional Sales data warehouse. An SSAS cube was developed to enable multidimensional analysis, and OLAP operations were performed using Excel to explore the data at different levels.

Power BI was used to create interactive reports, including matrix, slicers, drill-down, and drill-through reports, providing meaningful insights from the data. Overall, the assignment highlights how data warehousing and BI tools can be used together to support effective decision-making.